

# News, Price Volatility and Market Activity in Major Cryptocurrencies

Data Management course — Università degli Studi Milano-Bicocca (AY 2025/2026)

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Cryptocurrency markets are highly dynamic and volatile, often reacting to new information within very short time windows, while news coverage may also respond to sharp price movements.

This project investigates the timing of this interaction at hourly frequency by integrating Binance market data with GDELT news signals for Bitcoin (BTC), Ethereum (ETH), Binance Coin (BNB), Solana (SOL), and Ripple (XRP).

The analysis focuses on the relationship between news coverage, price volatility, and market activity, and is structured around four research questions to assess bidirectional lead–lag dynamics between news spikes and market behaviour.

## **RQ1. News spikes and subsequent price movement**

Do hours with a spike in news coverage coincide with larger price movement in the following horizons (1h, 6h, 24h)?

## **RQ2. Price movement preceding news spikes**

Are news spikes preceded by higher price volatility in the previous horizons (1h, 6h, 24h), suggesting that coverage reacts to ongoing moves?

## **RQ3. News spikes and subsequent market activity**

Do news spikes correspond to increased market activity, proxied by the number of trades, in the subsequent horizons (1h, 6h, 24h)?

## **RQ4. Price spikes and subsequent news coverage**

Do extreme price movements increase subsequent news coverage (future news volume and probability of at least one news spike within 24h)?

Hourly data were collected for five major cryptocurrencies over a **two-year window (Jan 25, 2024 – Jan 24, 2026)**, ensuring a consistent **1-hour granularity** and full alignment on a common UTC timeline.

- **News signals (GDELT DOC 2.0 – timeline API):** extracted in **weekly sub-windows** to preserve hourly resolution, using **asset-specific keyword queries** to retrieve **hourly article counts** related to each cryptocurrency (without downloading full article lists).
- **Market data (Binance Spot REST API – klines 1h):** retrieved via the **klines endpoint** with **time-based pagination** across the full period to collect hourly candles for each asset.

### *Collected variables*

Trading pair (symbol), open\_time, close\_time,  
open, high, low, close, n\_trades.

asset, ts\_hour\_utc, news\_count\_hour,  
avg\_tone\_hour, news\_share\_hour.

### 3. PRE-INTEGRATION DATA QUALITY

#### Section 3

**Uniqueness:** no duplicates for the composite key (asset, ts\_hour\_utc) in either source.

**Completeness:** Binance shows full coverage (17,520 hourly records per asset), with one extra boundary candle due to the inclusive window. GDELТ shows 16,981 hours per asset (coverage 0.969), with 539 missing hourly buckets per asset.

**Consistency:** prices are strictly positive,  $\text{high} \geq \text{low}$  always holds, and trading volumes are non-negative. For news sentiment, tone is treated as undefined when `news_count_hour = 0` and extreme tone outliers are filtered.

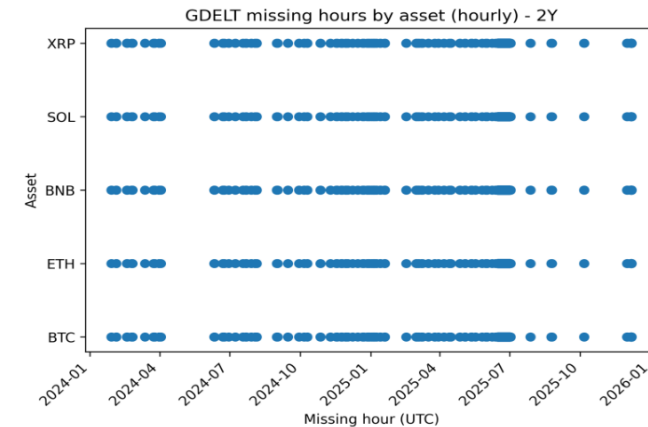


Figure 1- missing hours by asset

Timestamps were standardized to UTC and rounded down to the hourly bucket `ts_hour_utc`. Binance symbols were mapped to consistent asset identifiers (e.g., `BTCUSDT` → `BTC`).

Integration was performed by generating a complete hourly reference timeline for each asset, then applying left joins: first Binance market variables, then GDELT indicators. This guarantees a regular hourly panel while allowing GDELT to be missing when the endpoint does not return a bucket.

To separate true zero-news hours from endpoint-level gaps, a dedicated `gdelt_missing` flag was introduced. Missing `news_count_hour` values were imputed to zero for numeric stability, while `gdelt_missing` preserves missingness information for downstream analyses.

### Bincance Enrichment

Hourly price movement is captured through the continuously-compounded one-hour return `ret_1h`, central to the analysis for measuring the magnitude of market movements.

An intrahour dispersion proxy `hl_range_pct` is derived from the high–low range within the candle to measure within-hour volatility.

### GDELT Enrichment

A binary `news_spike` indicator is defined when `news_count_hour` exceeds an asset-specific 95th percentile threshold (`news_p95`) computed within each semester.

Semester-specific thresholds avoid over/under-detection when media intensity changes over time, preserving a stable notion of “unusually high attention”.

$$ret_{1h,t} = \ln \left( \frac{close_t}{close_{t-1}} \right) \text{ Where } close_t \text{ and } close_{t-1} \text{ are the log-transformed closing price.}$$

**The final dataset contains 87,600 rows and 16 columns (five assets × a complete two-year hourly panel).**

The curated dataset is stored in a relational database and queried in SQL. A relational model fits the problem because the data follow a fixed schema, are naturally keyed by (asset, ts\_hour\_utc), and require analytical operations based on filtering, grouping and time-window logic.

The table (enriched\_hourly\_2y) stores one record per asset–timestamp pair and enforces the composite key as a primary key to prevent duplicates and guarantee one observation per bucket.

All variables required by the analyses live in a single table, so research-question queries can be executed without additional joins.

All analyses are executed in SQL on `enriched_hourly_2y` using window functions, with a consistent event-comparison framework across the four research questions.

For each asset, lead and lag horizons of 1 hour, 6 hours and 24 hours are computed by ordering observations over `ts_hour_utc` and applying rolling frames on subsequent or previous rows. Multiple horizons are evaluated because market and media reactions may be delayed and may persist beyond the first hour.

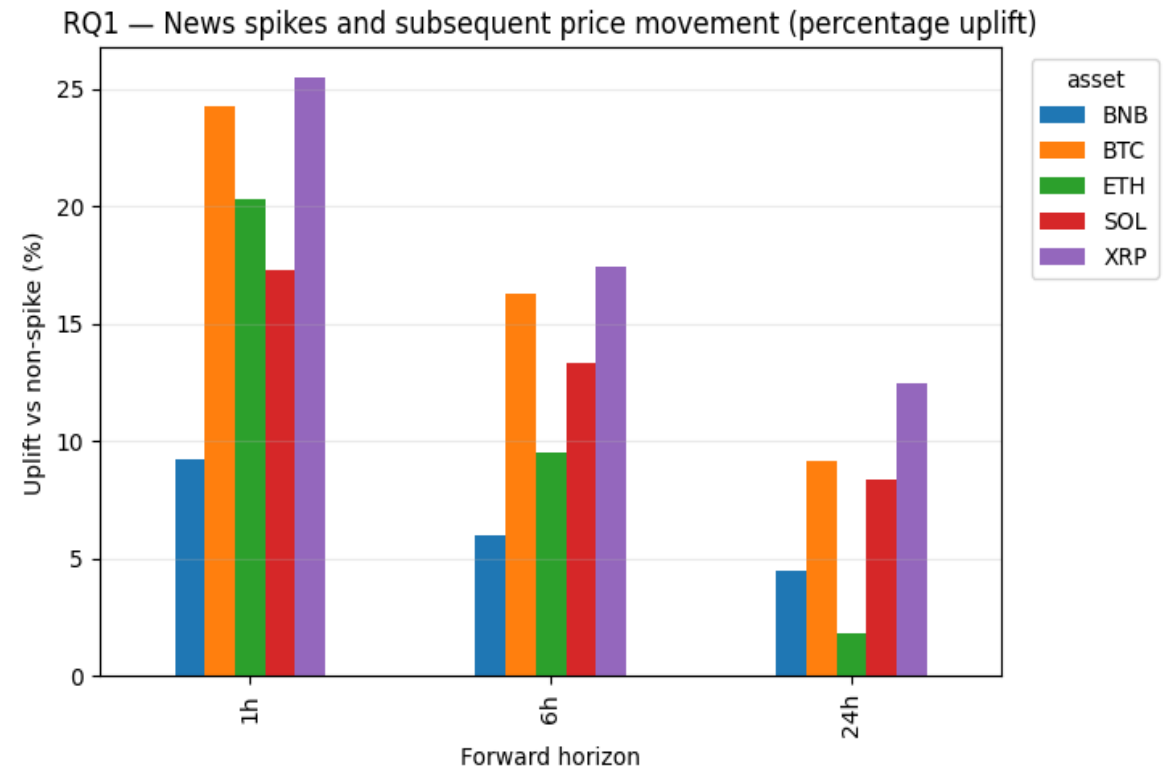
Horizon completeness is enforced (e.g., requiring all 24 forward rows to exist), and news-related computations are restricted to `gdelt_missing = 0` to avoid endpoint-level gaps biasing aggregates and probabilities.

## 7.1 RQ1. News spikes and subsequent price movement

RQ1

Across all five assets, news\_spike hours are followed by higher average absolute price movement at 1h, 6h and 24h horizons.

The strongest relative effects occur at the 1h horizon (BTC +24.2%, XRP +25.5%; ETH +20.3%, SOL +17.3%; BNB +9.3%). The uplift remains positive at 6h and 24h but tends to attenuate, indicating that the impact is most concentrated in the first hours after the spike.

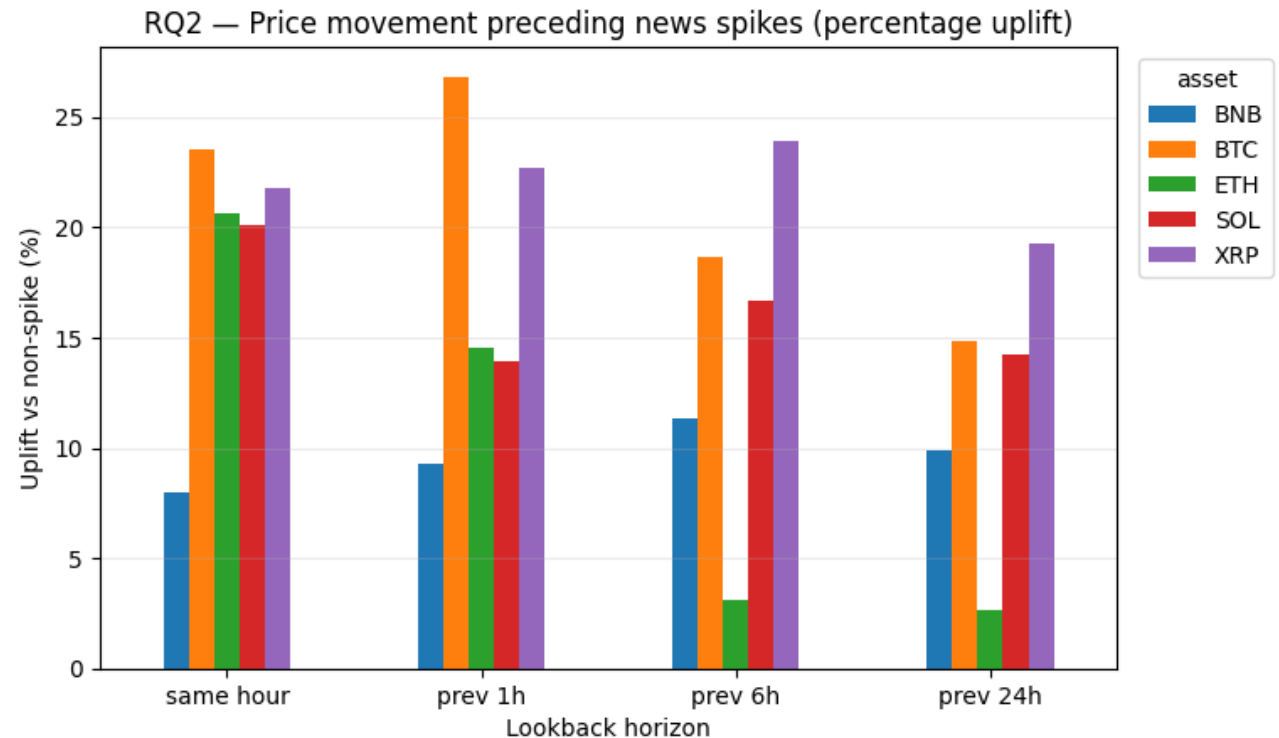


## 7.2 RQ2. News spikes and previous price movements

RQ2

News spikes generally occur during periods of already elevated price movement.

The pre-spike build-up is most pronounced for XRP and SOL at longer lookback windows (6h and 24h), while BTC shows the clearest gap already at the 1h lookback. ETH remains comparatively flat as the window extends, suggesting weaker pre-spike dynamics for that asset.

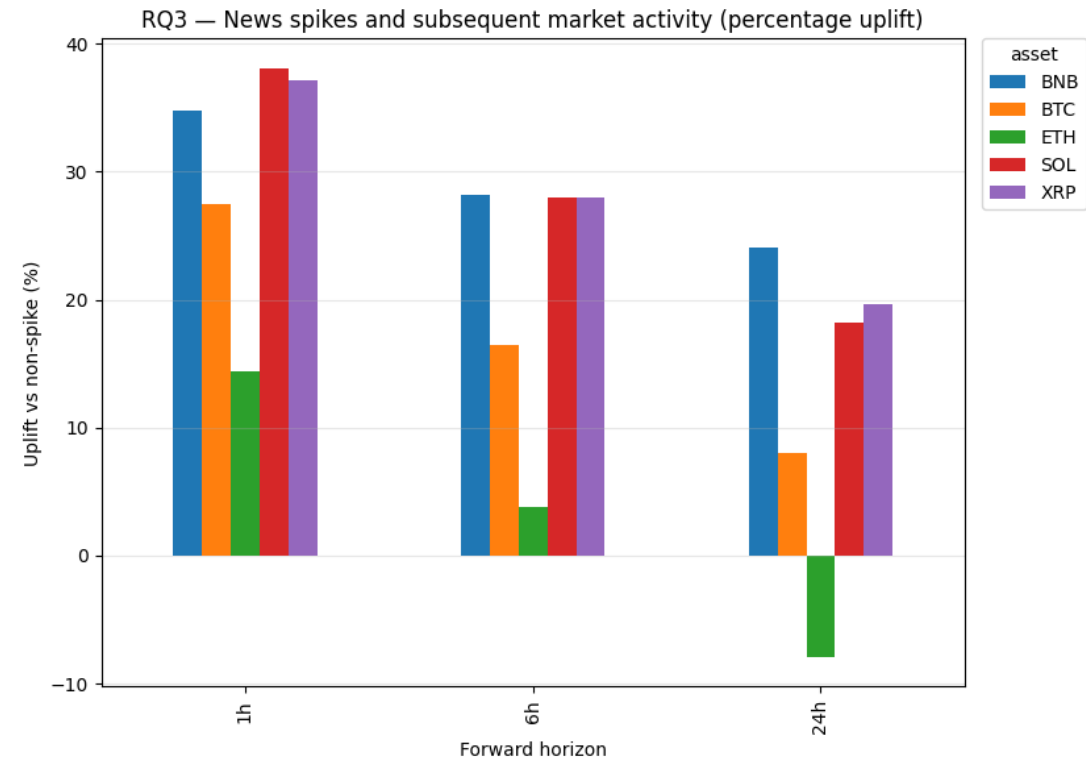


## 7.3 RQ3. News spikes and subsequent market activity

RQ3

News-spike hours are followed by higher trading activity at short horizons for all assets, with the largest 1h uplifts for SOL (~+38%) and XRP (~+37%), followed by BNB (~+35%) and BTC (~+27%).

At 6h the effect remains sizable for BNB, SOL and XRP, moderate for BTC, and small for ETH. Over 24h the uplift persists for four assets, while ETH becomes the main exception with a negative daily effect.

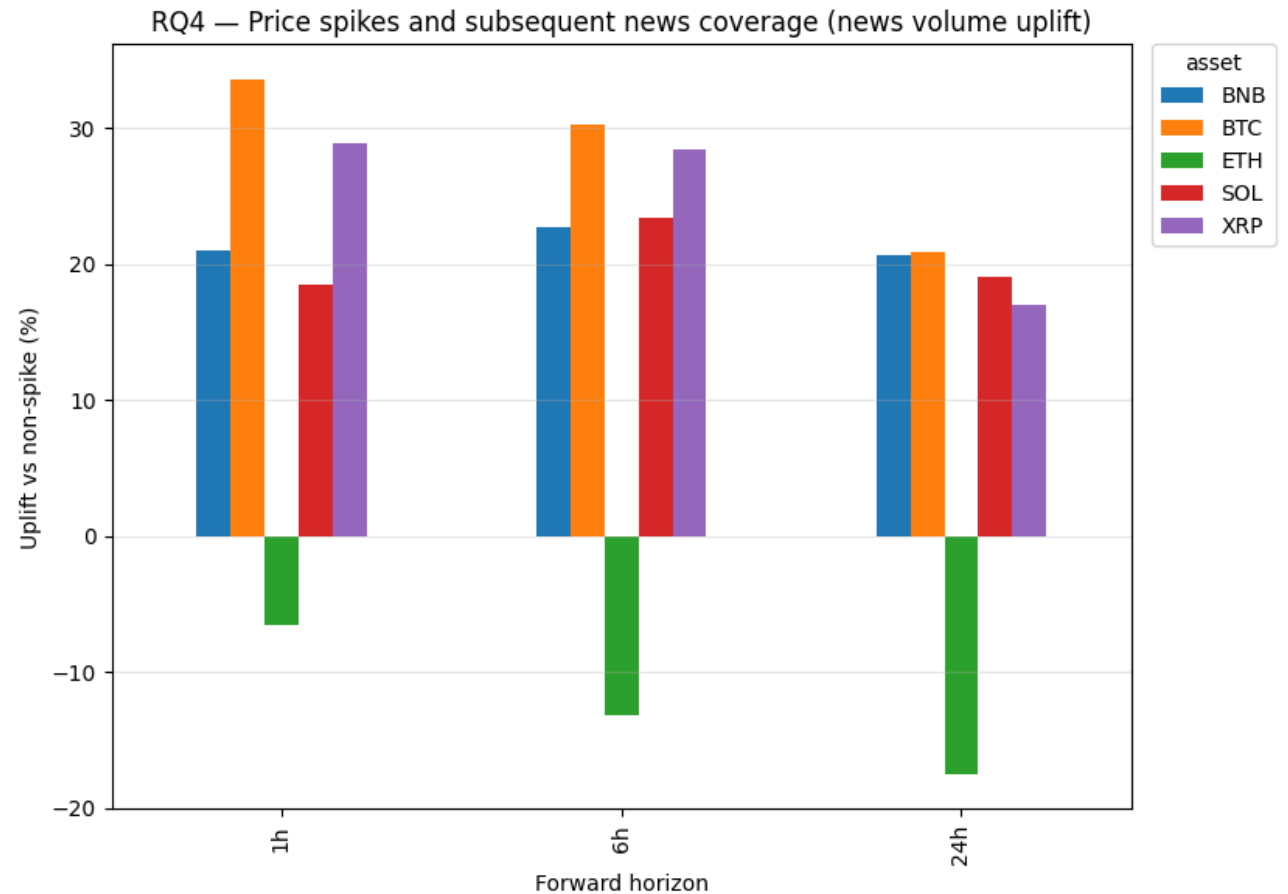


## 7.4 RQ4. Price spikes and subsequent news coverage

RQ4

Extreme price movements are followed by higher subsequent news coverage and a higher probability of observing at least one news spike within 24 hours for BTC, XRP, SOL and BNB.

BTC and SOL show the strongest post-event increases at longer horizons; XRP and BNB follow the same pattern. ETH is the main exception: future news volume decreases at all horizons, although the probability of a subsequent news spike still increases slightly.



This project built an end-to-end data management workflow: API-based acquisition (Binance + GDELT), pre-integration quality checks, UTC-based integration into a complete hourly panel, targeted enrichment (returns and spike indicators), relational storage, and SQL window queries.

Results indicate a bidirectional short-horizon relationship between news and market behaviour. News spikes are followed by higher price movement and trading activity across assets, with effects strongest in the first 1–6 hours (most pronounced for BTC, XRP, SOL, and BNB) and generally weakening by 24 hours, especially for ETH. Conversely, elevated volatility often precedes spikes, and price shocks tend to be followed by higher news volume and a higher probability of a news spike within 24 hours for most assets, with ETH showing the weakest or most mixed longer-horizon patterns.

**Limitations:** the study is descriptive and does not claim causality; GDELT query-based indicators can miss relevant coverage or include partial matches; fixed horizons (1h/6h/24h) simplify potentially richer lag structures.

**Future work:** add contextual market variables, move to statistical modeling, and estimate lag structures directly (e.g., adstock-style decaying exposure effects).